

Fig. 1.7 Illustration of the alanine dipeptide molecule. The central dihedral angles Φ and Ψ are shown which are relevant to describing the conformational states

When, as in the case of the alanine dipeptide, net charges are present on the atoms, one may model this by means of Coulomb potentials:

$$\varphi_{ij}^{\text{Coulomb}}(r_{ij}) = \frac{CQ_iQ_j}{\epsilon r_{ij}},$$

where $\epsilon > 0$ is the dielectric constant, and Q_i, Q_j are the charges on atoms i and j, respectively. C is a positive coefficient allowing the adjustment of units. Obviously the effect of the Coulombic potential depends strongly on whether the atoms have the same or oppositely signed charges.

In the simplest treatments, the Coulomb potential is simply cut off at distance r_{cut}, i.e., is taken to be zero for $r > r_{\text{cut}}$. This should be done in such a way that the potential remains at least continuously differentiable, preferably smoother (see Exercise 11).[2] However simply neglecting the potential energy after some distance r_{cut} (i.e. introducing a discontinuity in the potential energy function) can be catastrophic in larger systems where long potential energy tails play a significant role (see [242], for instance, where using a cut-off gives nearly an order-of-magnitude error in the computed electrostatic energy). More accurate treatments of the long range behavior include the use of an exponential term involving the Debye length κ which models screening due to the presence of a polar solvent, in which case the potential is modified to have the form of a *Yukawa potential*:

$$\varphi_{ij}^{\text{screened}}(r_{ij}) = \frac{CQ_iQ_j}{\epsilon r_{ij}} e^{-\kappa r_{ij}}.$$

The effect of screening is illustrated in Fig. 1.8. Further modifications substitute potential energies that smoothly go to a constant value (normally zero) at a finite distance, as, e.g. the damped, shifted force model [134].

In addition to length bonds, Lennard-Jones or other short range forces, and Coulomb potentials, it may also be necessary to incorporate *angle bonds* and

[2] In fact, many results and estimates given in this book technically require greater smoothness, but in practice discontinuity in second or higher derivatives of the potential at a single point does not usually cause great difficulty.